

Theory of Mechanism and Machine

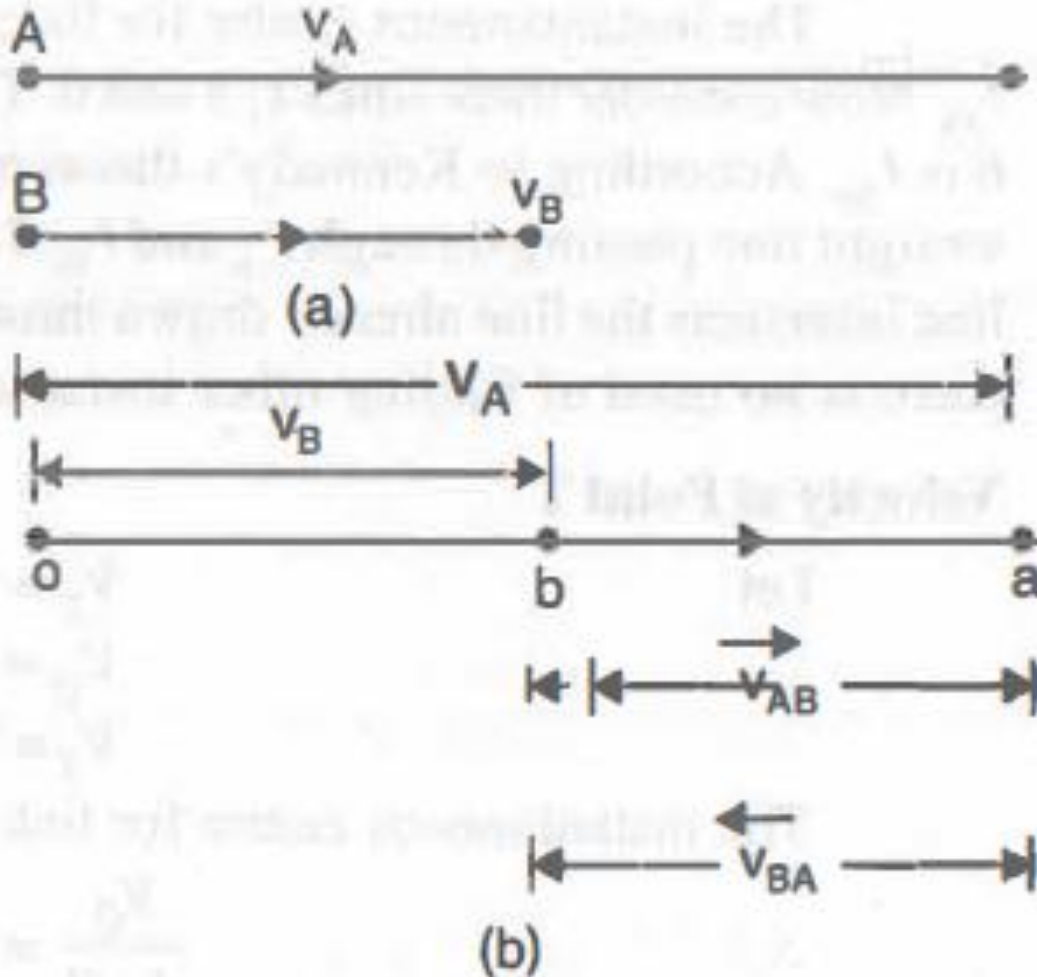
(Week-5)

Velocity Mechanism using Relative Velocity Approach:

- *The relative velocity method is very useful and rapid method for determining linear and angular velocities in the mechanism. This method has the additional advantage over instantaneous centre method that it can be used for making acceleration analysis.*
- *Before discussing this method, let us consider the following important cases:*
 - *(i) Relative velocity of two bodies moving in straight lines.*
 - *(ii) Relative velocity of any point on a link with respect to another point.*

Relative velocity of two bodies moving in straight lines:

The Figure shows the two bodies A and B which are moving along parallel lines in the same direction.



v_{AB} = Vector difference of v_A and v_B

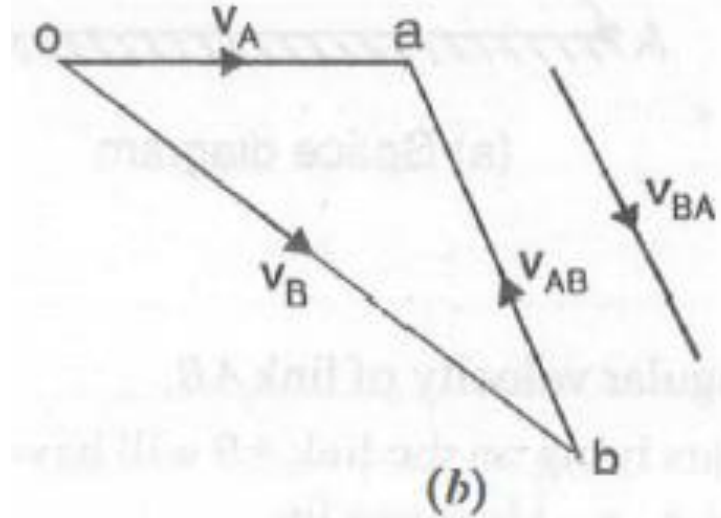
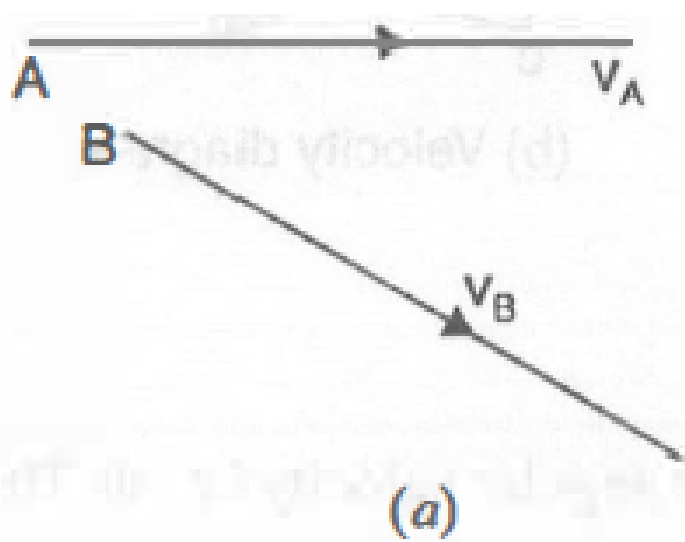
$$= \vec{v}_A - \vec{v}_B$$

v_{BA} = Vector difference of v_B and v_A

$$= \vec{v}_B - \vec{v}_A$$

Relative Velocity of Two Bodies Moving along Inclined lines:

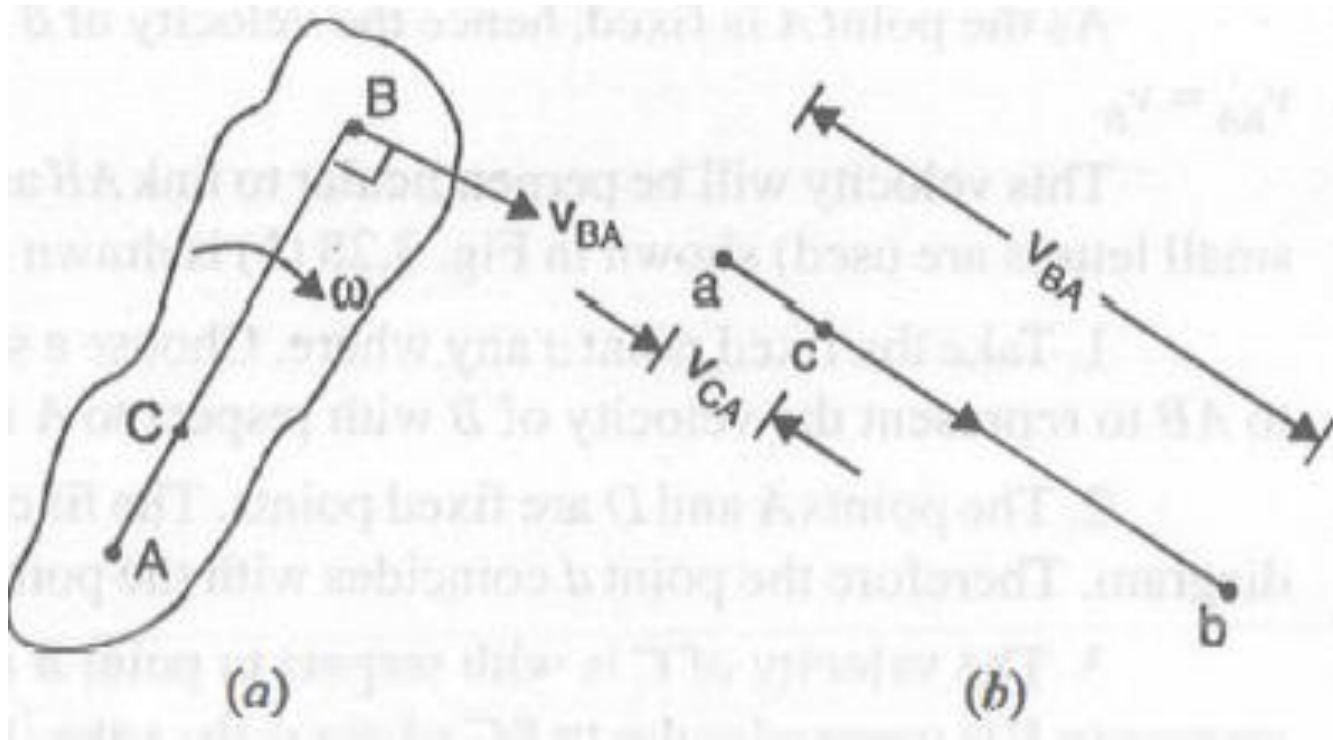
The Figure. shows the two bodies *A* and *B* moving along inclined lines with absolute velocities v_A and v_B respectively.



- Then relative velocity of *A* with respect to *B* is written as v_{AB} and is given by, $v_{AB} = \text{Vector } ba$
- The relative velocity of *B* with respect to *A* is given by, $v_{BA} = \text{Vector } ab$

Relative Velocity of a Rigid Link:

AB is a rigid link. In case of a rigid link the velocity of any point on the link with respect to another point on the same link is always perpendicular to the line joining these points on the space diagram. Hence the velocity of B with respect to A will be perpendicular to AB as shown in Fig. (a).



- The Fig. (b) shows the velocity diagram for the rigid link AB. The relative velocity of B with respect to A (i.e., v_{BA}) is represented by the vector ab which is perpendicular to the line AB.

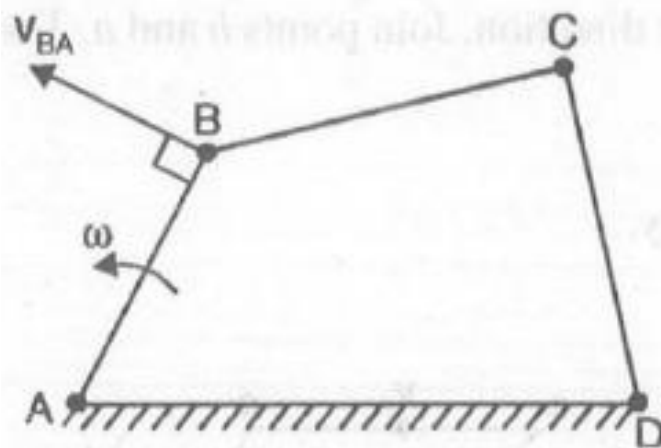
$$v_{BA} = \vec{ab} = \omega \times AB$$

$$v_{CA} = \vec{ac} = \omega \times AC$$

$$\frac{v_{CA}}{v_{BA}} = \frac{\vec{ac}}{\vec{ab}} = \frac{\omega \cdot AC}{\omega \cdot AB} = \frac{AC}{AB}.$$

Velocities in Four Bar Chain:

A four bar chain ABCD is shown in Fig (a). The link AD is fixed and link AB is rotating anti-clockwise with a known angular velocity.



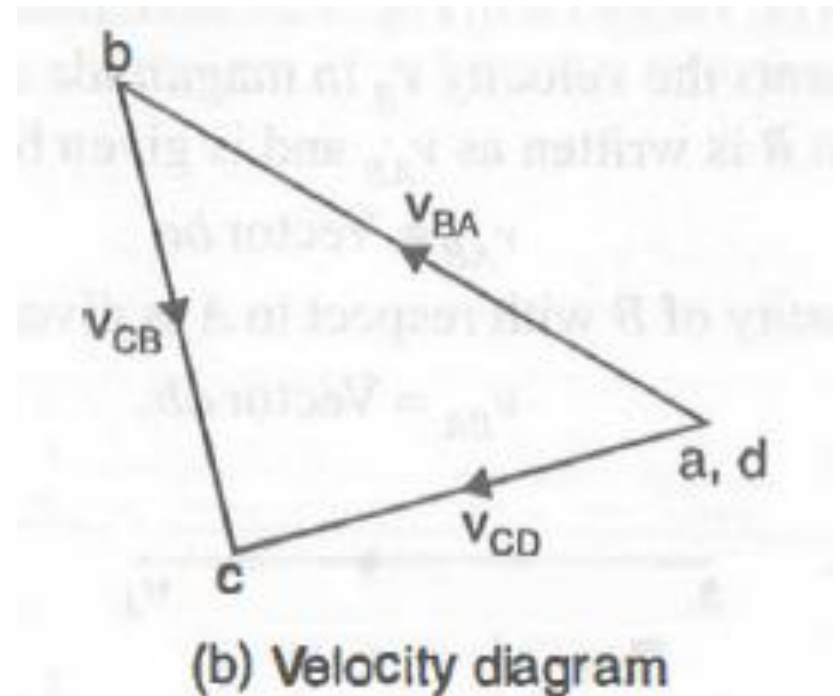
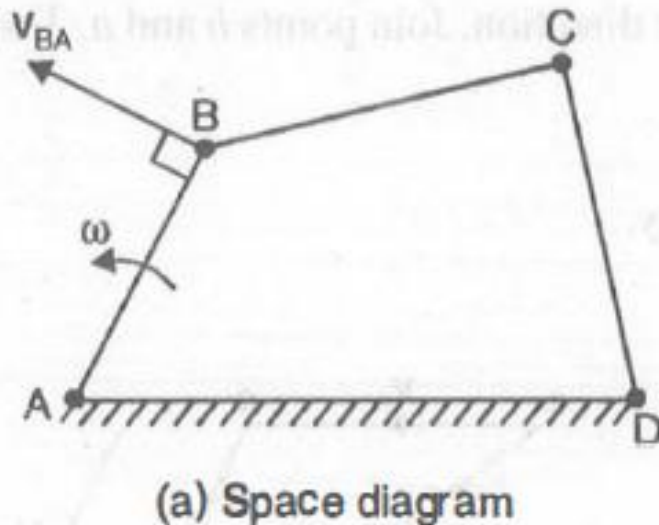
(a) Space diagram

ω = Angular velocity of link AB.

The linear velocity of B with respect to A (v_{BA}) is given by

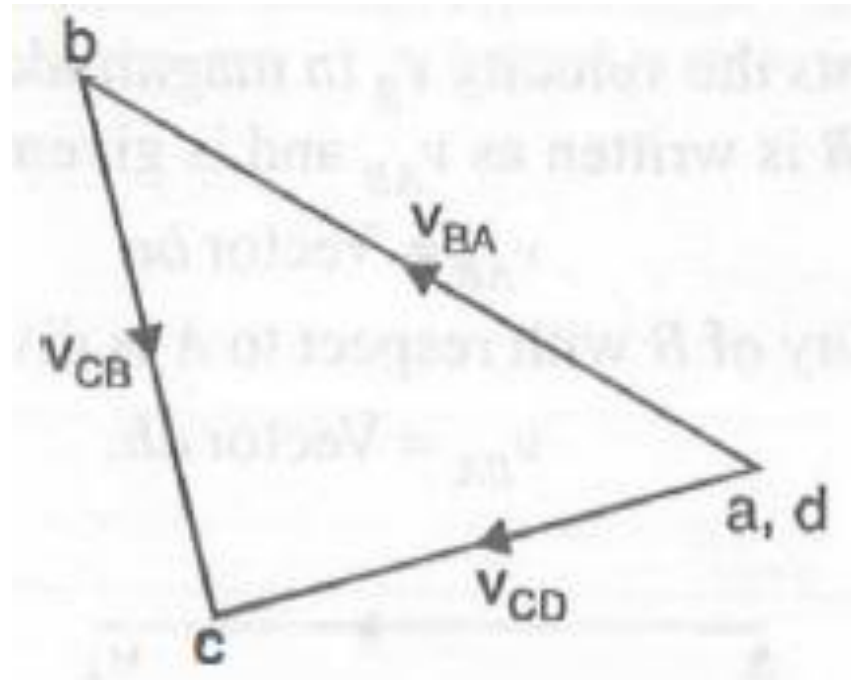
$$v_{BA} = \text{Angular velocity of link AB} \times \text{Length AB} = \omega \times AB$$

- As the point A is fixed, hence the velocity of B with respect to A will be equal to velocity of B also i.e. $V_{BA} = V_B$.
- This velocity will be perpendicular to link AB as shown in Fig. (a). The velocity diagram (in which small letters are used) shown in Fig. (b) is drawn as discussed below:



- 1 . Take the fixed point 'a' any where. Choose a suitable velocity scale and draw vector ab perpendicular to AB to represent the velocity of B with respect to A i.e., V_{BA}
2. The points A and D are fixed points. The fixed points of the mechanism are coincident in the velocity diagram. Therefore the point 'd' coincides with the point 'a' in the velocity diagram.
3. The velocity of C is with respect to point B and also with respect to point D. The velocity of C with respect to B is perpendicular to BC whereas the velocity of C with respect to D is perpendicular to CD. From 'b' draw vector be perpendicular to BC to represent the velocity of C with respect to B i.e. V_{CB} .
4. Also from 'a' or 'd' draw vector dc perpendicular to DC to represent the velocity of C with respect to D i. e., V_{CD} . The vectors be and de (or ae) intersect at e.

Then *abc* represents the velocity diagram for the given four bar chain.



$$\text{Angular velocity of } BC = \frac{\text{Vector } bc}{\text{Length } BC}$$

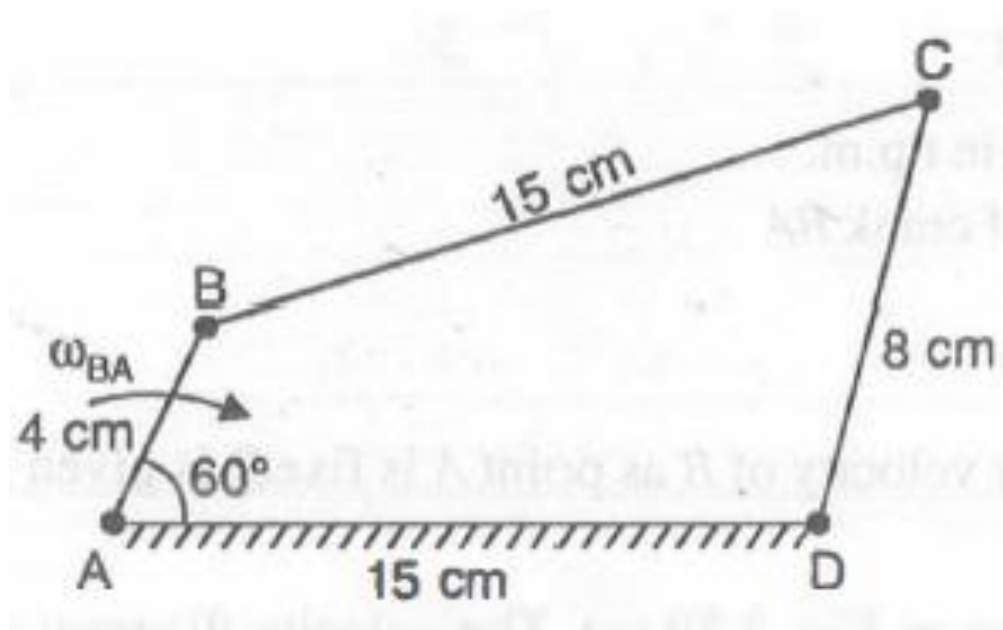
$$\text{and angular velocity of } CD = \frac{\text{Vector } cd}{\text{Length } CD}$$

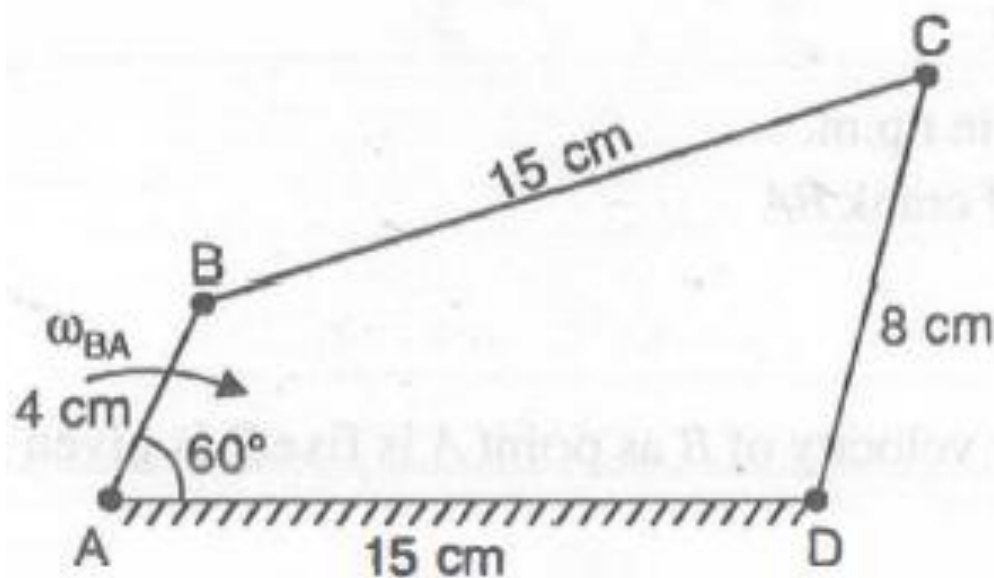
Note. (i) For velocity diagram small letters are used.

(ii) For space diagram capital letters are used.

Q 1: In a four bar chain ABCD, AD is fixed and is 15 cm long. The crank AB is 4 cm long and rotates at 120 r.p.m. clockwise, while the link CD=8 cm oscillates about D. BC and AD are of equal length. Find the angular velocity of link CD when angle BAD=60°.

Draw the FBD as per the given data:





AD is fixed and $= 15$ cm, Crank $AB = 4$ cm

Speed of crank AB , $N_{BA} = 120$ r.p.m.

Link $CD = 8$ cm, $BC = AD = 15$ cm and angle $BAD = 60^\circ$

Now angular velocity of the link AB ,

$$\omega_{BA} = \frac{2\pi N_{BA}}{60} = \frac{2\pi \times 120}{60} = 12.56 \text{ rad/s}$$

∴ Velocity of B with respect to A (i.e., v_{BA}) or simply velocity of B as A is fixed (i.e., v_B) is given by

$$v_{BA} \text{ (or } v_B) = \omega_{BA} \times \text{Length } AB = 12.56 \times 4 = 50.24 \text{ cm/s}$$

The direction of this velocity is perpendicular to AB .

Velocity Diagram

Take the fixed point 'a' anywhere. Choose a suitable scale and draw vector ab perpendicular to AB to represent the velocity of B with respect to A (or simply the velocity of B) i.e., v_{BA} (v_B) such that vector $ab = v_{BA}$ or $v_B = 50.24$ cm/s

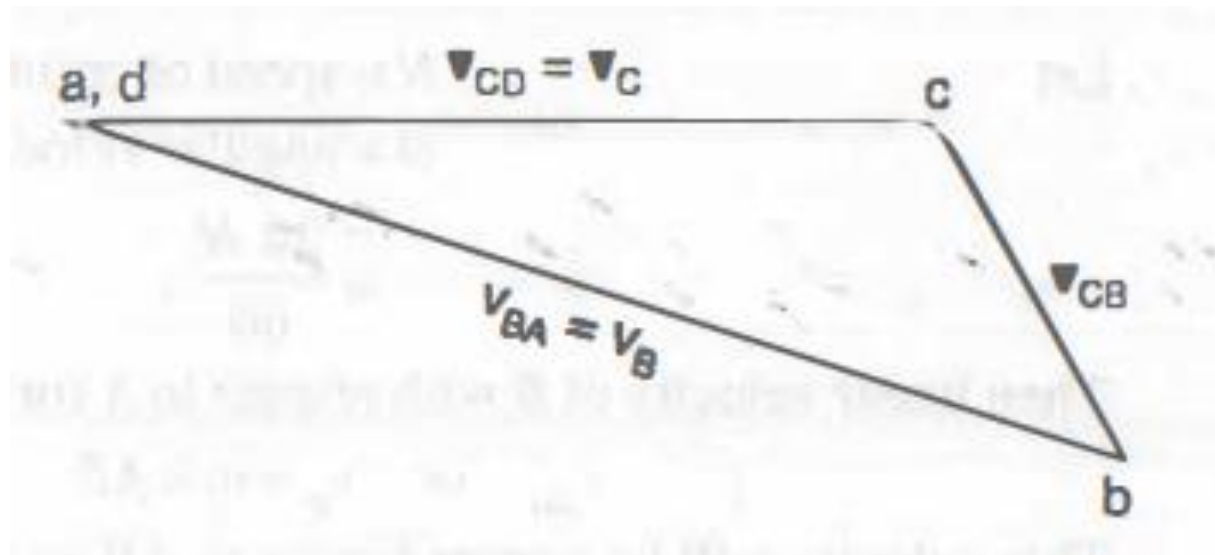
Since points A and D are fixed hence the corresponding points 'a' and 'd' in velocity diagram will coincide.

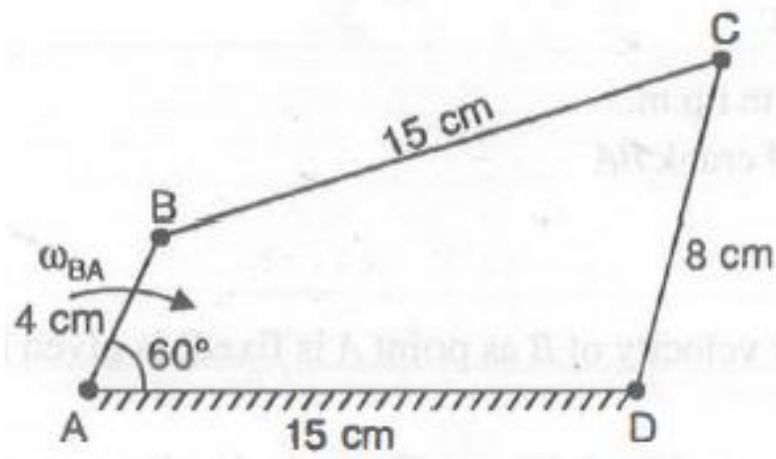
Now the point C moves with respect to point B and also with respect to D . The velocity of C with respect to B is perpendicular to BC whereas the velocity of C with respect to D is perpendicular to CD

Hence from point b , draw vector be perpendicular to BC to represent the velocity of C with respect to B (V_{CB})

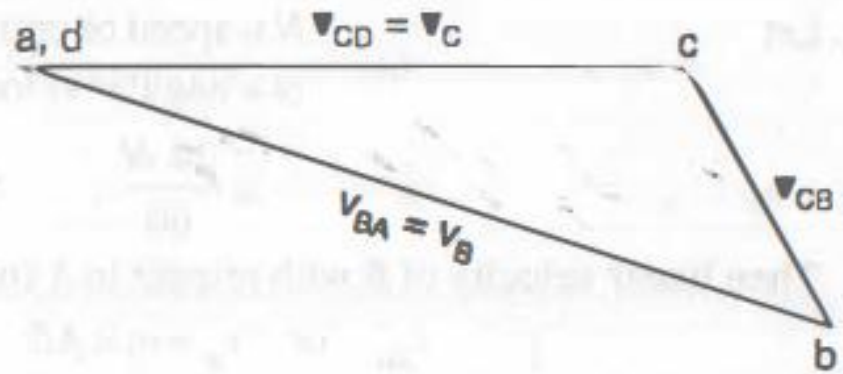
From point, d (or a) draw vector dc (or ac) perpendicular to DC to represent the velocity of C with respect to D (V_{CD}) or simply velocity of C i.e., V_C (as point D is fixed).

The vectors be and de (or ac) intersect at e . Then abc is velocity triangle.





(a) Space diagram



(b) Velocity diagram

On measurement from velocity triangle, velocity of C with respect to D is given by,

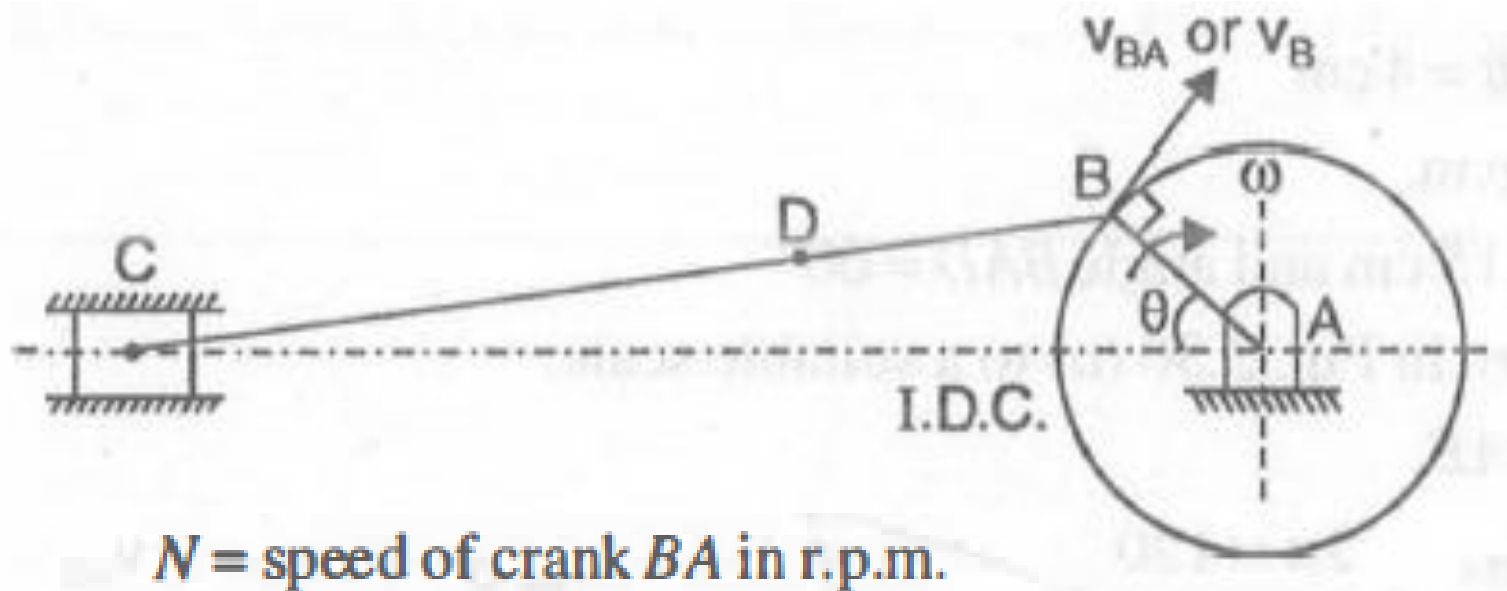
$$v_{CD} = \text{vector } dc \text{ (or } ac) = 38 \text{ cm/s.}$$

$\therefore$ Angular velocity of link CD is given by,

$$\omega_{CD} = \frac{v_{CD}}{\text{Length } CD} = \frac{38}{8} = 4.75 \text{ rad/s (clockwise). Ans.}$$

Velocities in Slider Crank Mechanism:

- The Fig. shows a slider crank mechanism, in which the crank AB is rotating clockwise with a given speed. The slider C is moving along the line CA. The connecting rod is represented by CB.*



N = speed of crank BA in r.p.m.

ω = angular velocity of crank BA

$$= \frac{2\pi N}{60}.$$

This velocity will be perpendicular to AB as shown in Fig.

Then linear velocity of B with respect to A (or only velocity of B as point A is fixed) is given by,

$$v_{BA} \quad \text{or} \quad v_B = \omega \times AB$$

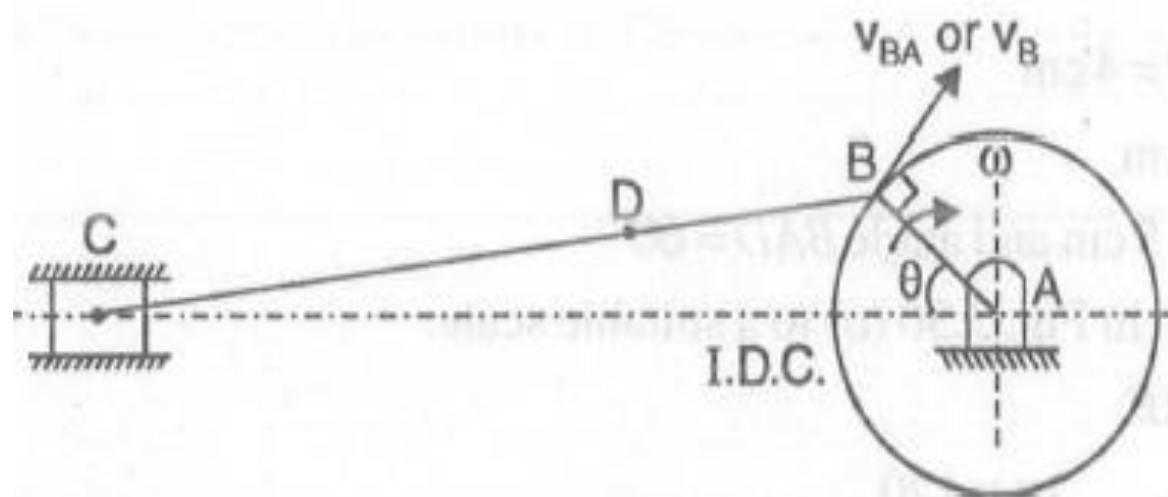
Velocity Diagram

1) Take the fixed point 'a' anywhere. Choose a suitable velocity scale and draw vector 'ab' perpendicular to 'AB' to represent the velocity of 'B' w.r.t to 'A' i.e. $v_{BA} = ab$ vector.

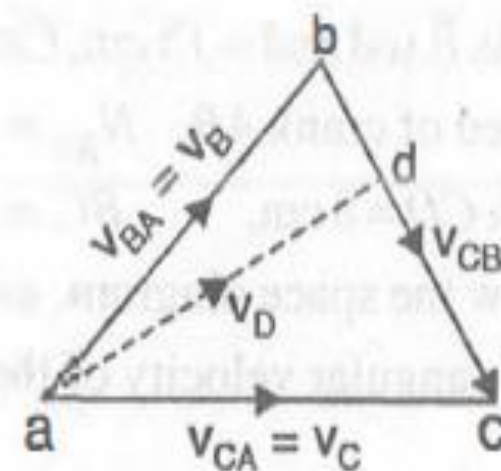
2) The velocity of 'C' is w.r.t 'B' and also 'A'. The velocity of 'C' w.r.t to 'B' is perpendicular to BC, hence from 'b' draw vector bc perpendicular to BC i.e. v_{CB} .

3) While the velocity of 'C' w.r.t 'A' is in horizontal direction along CA. Draw a line from point 'a' horizontally i.e. vector ac.

4) The vectors bc and ac intersect at c. Then abc represents the velocity diagram.



(a) Space diagram



(b) Velocity diagram

3. The absolute velocity of any point D on the connecting rod BC can be obtained by dividing vector bc such that

$$\frac{bd}{bc} = \frac{BD}{BC}$$

$$\therefore bd = \frac{BD}{BC} \times bc$$

As BD , BC and vector bc are known, hence bd can be calculated. Then location of point d on vector bc is obtained. Now join point a to d . Then vector ad represents in magnitude and direction the absolute velocity of the point D .

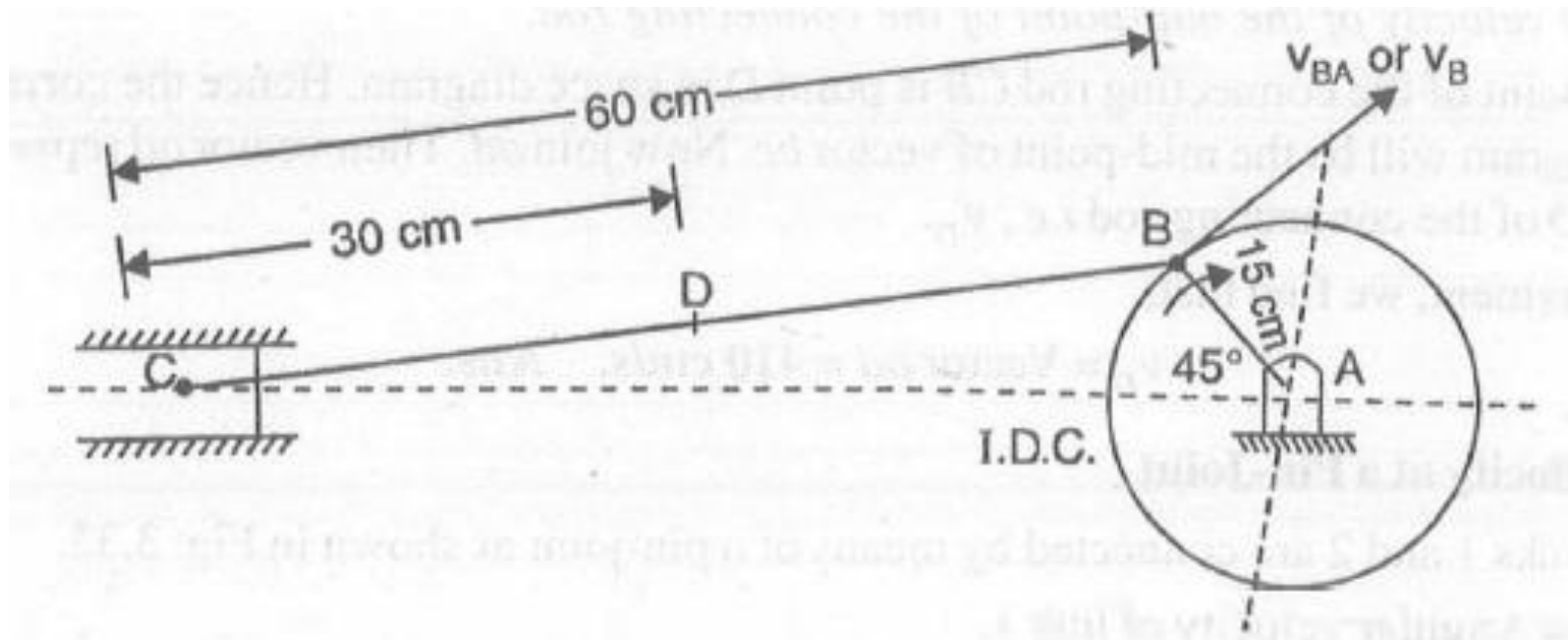
- Q 2: The crank of a slider crank mechanism is 15 cm and the connecting rod is 60 cm long. The crank makes 300 r.p.m. in the clockwise direction. When it has turned 45° from the inner dead centre position, determine :

(i) Velocity of slider C,

(ii) Angular velocity of connecting rod and

(iii) Linear velocity of the mid-point of the connecting rod.

Draw the FBD:



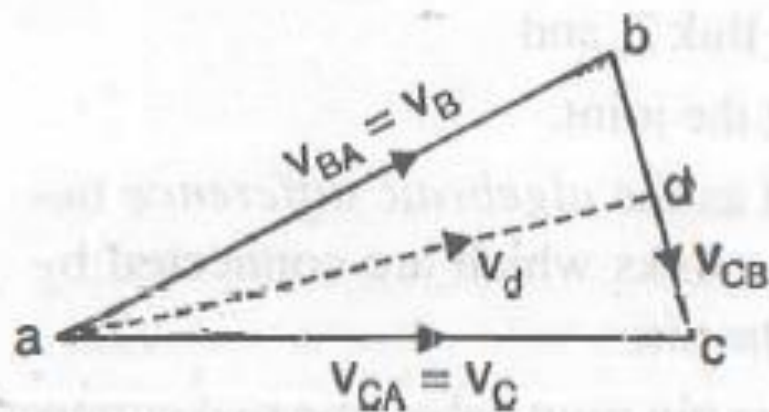
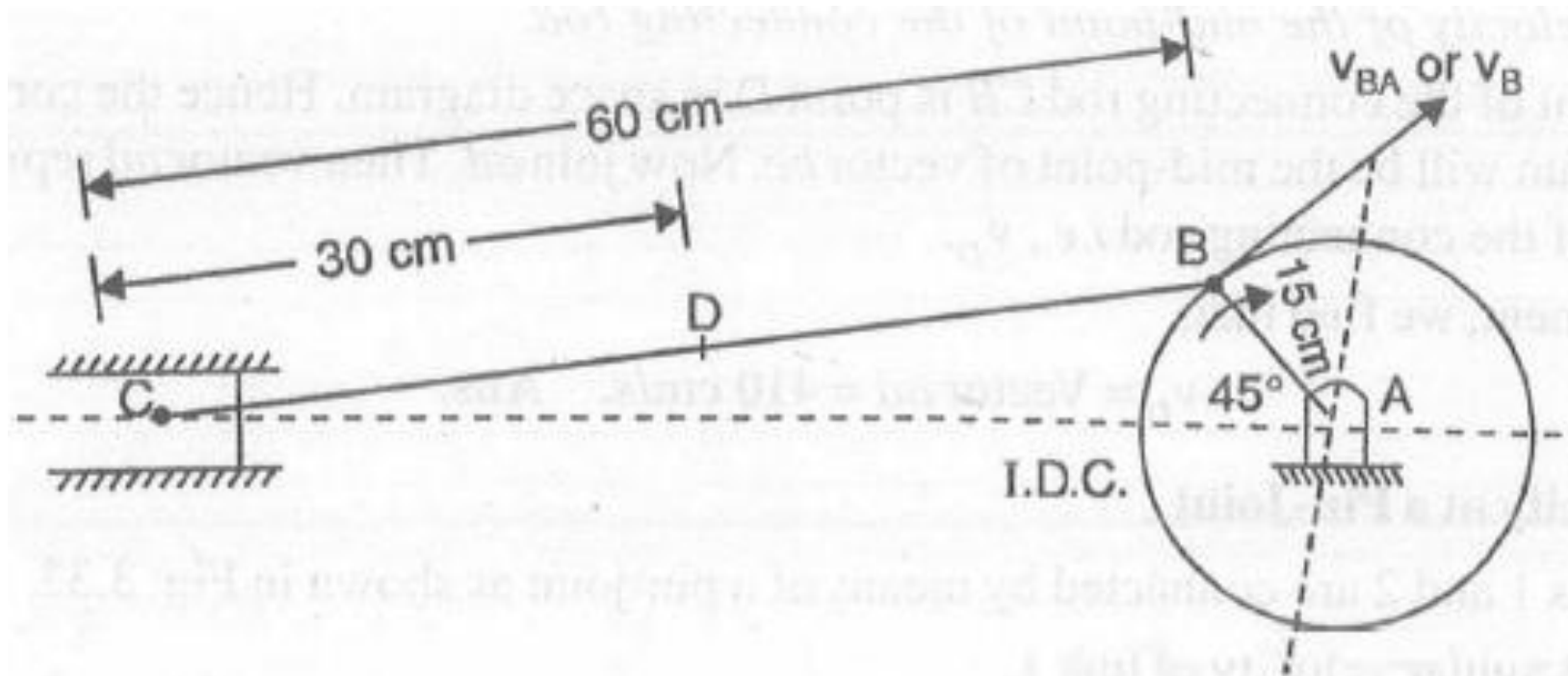
Now angular velocity of crank is given by,

$$\omega = \frac{2\pi \times N}{60} = \frac{2\pi \times 300}{60} = 31.42 \text{ rad/s.}$$

$\therefore$ Linear velocity of B with respect to A (or velocity of B) is given by,

$$v_{BA} = v_B = \omega \times AB = 31.42 \times 15 = 471.3 \text{ cm/s}$$

The direction of v_{BA} will be at right angles to AB .



(a) Space diagram

(b) Velocity diagram

By measurement, we find velocity of C with respect to B ,

$$v_{CB} = \text{Vector } bc = 340 \text{ cm/s}$$

and velocity of C with respect to A or simply velocity of C ,

$$v_{CA} \text{ or } v_C = \text{Vector } ac = 400 \text{ cm/s.}$$

(i) $\therefore$ Velocity of slider $C = v_C = 400 \text{ cm/s.}$ Ans.

(ii) Angular velocity of connecting rod CB is given by,

$$\omega_{CB} = \frac{v_{CB}}{\text{Length } BC} = \frac{340}{60} = 5.67 \text{ rad/s.}$$

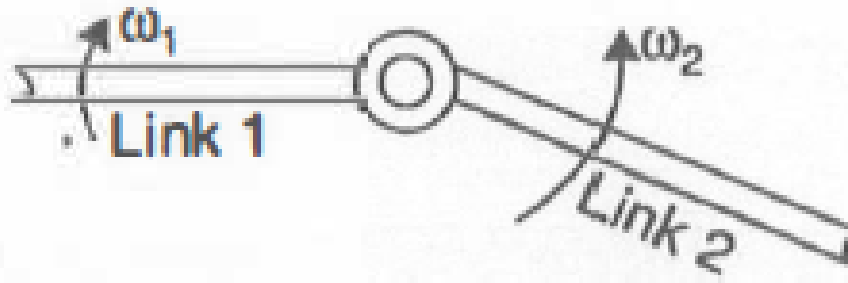
(iii) *Linear velocity of the mid-point of the connecting rod.*

The mid-point of the connecting rod CB is point D in space diagram. Hence the corresponding point d in the velocity diagram will be the mid-point of vector be . Now join ad . Then vector ad represents the velocity of the mid-point D of the connecting rod i.e., v_D .

By measurement, we find that,

$$v_D = \text{Vector } ad = 410 \text{ cm/s.}$$

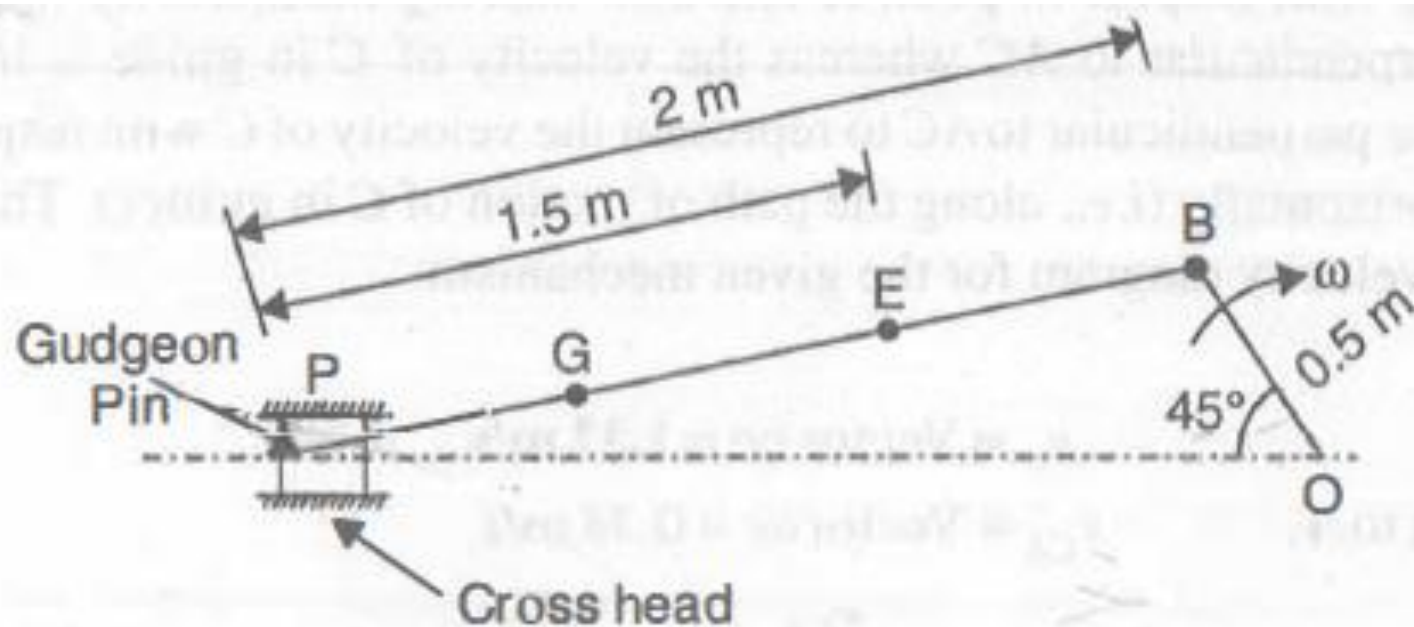
- **Rubbing Velocity at a Pin-Joint:**
- The two links 1 and 2 are connected by means of a pin-joint as shown in Fig.



- Let ω_1 = Angular velocity of link 1
- ω_2 = Angular velocity of link 2
- r = Radius of the pin at the joint.
- The rubbing velocity at the pin joint, when the two connected links move in opposite direction is $r(\omega_1 + \omega_2)$.
- The rubbing velocity at the pin joint, when the two connected links move in same direction is $r(\omega_1 - \omega_2)$.
- If a pin connects one sliding member and the other turning member the angular velocity of the sliding member is zero and hence the velocity of rubbing will be given by $r\omega$

Q 3: The crank and connecting rod of a horizontal steam engine are 0.5 m and 2 m long respectively. The crank makes 180 r.p.m. in the clockwise direction. When it has turned 45° from the inner dead centre position, determine :

- (i) Velocity of piston,*
- (ii) Angular velocity of connecting rod,*
- (iii) Velocity of point E on connecting rod 1.5 m from the gudgeon pin,*
- (iv) Velocity of rubbing at the pins of the crank shaft, crank and cross-head when the diameters of their pins are 5 cm, 3 cm, 6 cm respectively,*
- (v) Position and linear velocity of any point G on the connecting rod which has the least velocity relative to crank-shaft.*



Crank length, $OB = 0.5 \text{ m}$

Connecting rod length, $BP = 2 \text{ m}$

Crank speed = 180 r.p.m.

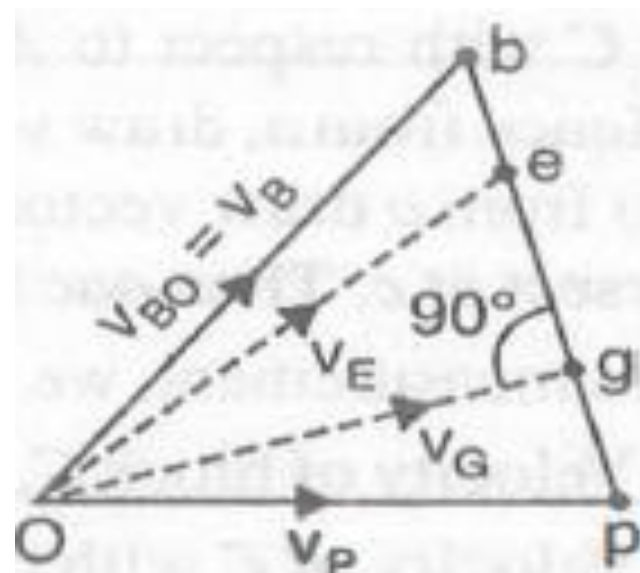
Angle turned by crank from inner dead centre = 45°

Now $N_{BO} = 180 \text{ r.p.m.} \quad \therefore \omega_{BO} = \frac{2\pi \times 180}{60} = 18.85 \text{ rad/s.}$

Hence the velocity of B with respect to O is given by,

$$v_{BO} = \omega_{BO} \times \text{Length } OB = 18.85 \times 0.5 = 9.425 \text{ m/s}$$

The direction of v_{BO} is perpendicular to BO .



1. Take any point o . Choose a suitable velocity scale and draw vector ob perpendicular to OB to represent the velocity of B with respect to O (i.e., v_{BO}) such that vector $ob = v_{BO} = 9.425$ m/s.

2. Now the point P is moving with respect to B and also with respect to O . The velocity of P with respect to point B is perpendicular to PB whereas the velocity of P with respect to O is along PO . Hence from point b , draw vector bp perpendicular to BP to represent the velocity of P with respect to B and from point o draw vector op parallel to the path of motion of P (which is along PO) to represent the velocity of P with respect to O or simply the velocity of P i.e., v_P . The vectors bp and op intersect at p .

3. The point E , which is at a distance of 1.5 m from P , will have a corresponding point e in the velocity diagram. The location of e in the velocity diagram is obtained by dividing the vector bp at e in the same ratio, as E divides PB in space diagram. This means

$$\frac{ep}{bp} = \frac{EP}{BP}$$

$$\therefore ep = \frac{EP}{BP} \times bp = \frac{1.5}{2.0} \times bp = 0.75bp$$

Hence location of e in vector bp can be determined. Now join o to e .

Hence location of e in vector bp can be determined. Now join o to e .

Then vector oe represent the velocity of E i.e., v_E .

(i) Velocity of piston (or point P),

$$v_P = \text{Vector } op = 8.15 \text{ m/s. Ans.}$$

(ii) Angular velocity of connecting rod BP ,

$$\omega_{BP} = \frac{v_{BP}}{BP}$$

But

$$v_{BP} = \text{Vector } pb = 6.8 \text{ m/s and } BP = 2 \text{ m}$$

$\therefore$

$$\omega_{BP} = \frac{6.8}{2} = 3.4 \text{ rad/s (clockwise). Ans.}$$

(iii) Velocity of point E ,

$$v_E = \text{Vector } oe = 8.5 \text{ m/s. Ans.}$$

(iv) Velocity of rubbing at the pins :

Dia. of crank shaft pin at O , $d_o = 5 \text{ cm}$

Dia. of crank pin at B , $d_b = 3 \text{ cm}$

Dia. of cross head pin at P , $d_c = 6 \text{ cm}$

Now velocity of rubbing at the pin of crank-shaft

$$= \frac{d_o}{2} \times \omega_{BO} = \frac{5}{2} \times 18.85 = 47.125 \text{ cm/s. Ans.}$$

Velocity of rubbing at the pin of the cross-head

$$= \frac{d_c}{2} \times \omega_{BP} = \frac{6}{2} \times 3.4 = 10.2 \text{ cm/s. Ans.}$$

Velocity of rubbing at the pin of crank (*i.e.*, at B)

$$= \frac{d_o}{2} (\omega_{BO} + \omega_{BP}) = \frac{3}{2} (18.85 + 3.4) = 33.375 \text{ cm/s.}$$

The position of point G on the connecting rod which has the least velocity relative to crank shaft, is determined by drawing perpendicular from point o to vector bp . Since the length og will be the least, hence point g represents the required position of G on the connecting rod. The position of G is obtained as

$$\frac{bg}{bp} = \frac{BG}{BP}$$

$$\therefore BG = \frac{bg}{bp} \times BP = \frac{5}{6.8} \times 2 = 1.47 \text{ m. Ans.}$$

($\because bg = 5$ and $bp = 6.8$ from velocity diagram)

Now linear velocity of G , $v_G = \text{Vector } og = 8 \text{ m/s. Ans.}$